Q1. Hospital Patient Records

Objective

Apply the **Apriori association rule mining algorithm** with the following constraints:

- Minimum Support = **50%**
- Minimum Confidence = **70%**
- Every generated rule must contain **Fever**

Dataset

Patient	Fever	Cough	Headache	Fatigue
P1	Yes	Yes	No	No
P2	Yes	No	Yes	No
P3	No	Yes	Yes	No
P4	Yes	Yes	Yes	No
P5	Yes	No	No	Yes

WEKA Implementation

The dataset was loaded into **WEKA Explorer → Associate → Apriori**.

The following parameters were used:

- Minimum support = **0.50**
- Minimum confidence = **0.70**
- Number of rules = **10**

WEKA Output Screenshot

[Insert your WEKA Apriori screenshot here]

Figure 1: Apriori association-rule mining output for the Hospital Patient Records dataset.

Valid Association Rule

From the WEKA output, the valid rule satisfying the constraints is:

Fever = Yes → Fatigue = No

Support:

$$Support = \frac{3}{5} \times 100 = 60\%$$

Confidence:

$$Confidence = \frac{3}{4} \times 100 = 75\%$$

Constraint Verification

The rule satisfies all the required constraints:

- **Support = 60% ≥ 50%**
- **Confidence = 75% ≥ 70%**
- The rule contains **Fever**

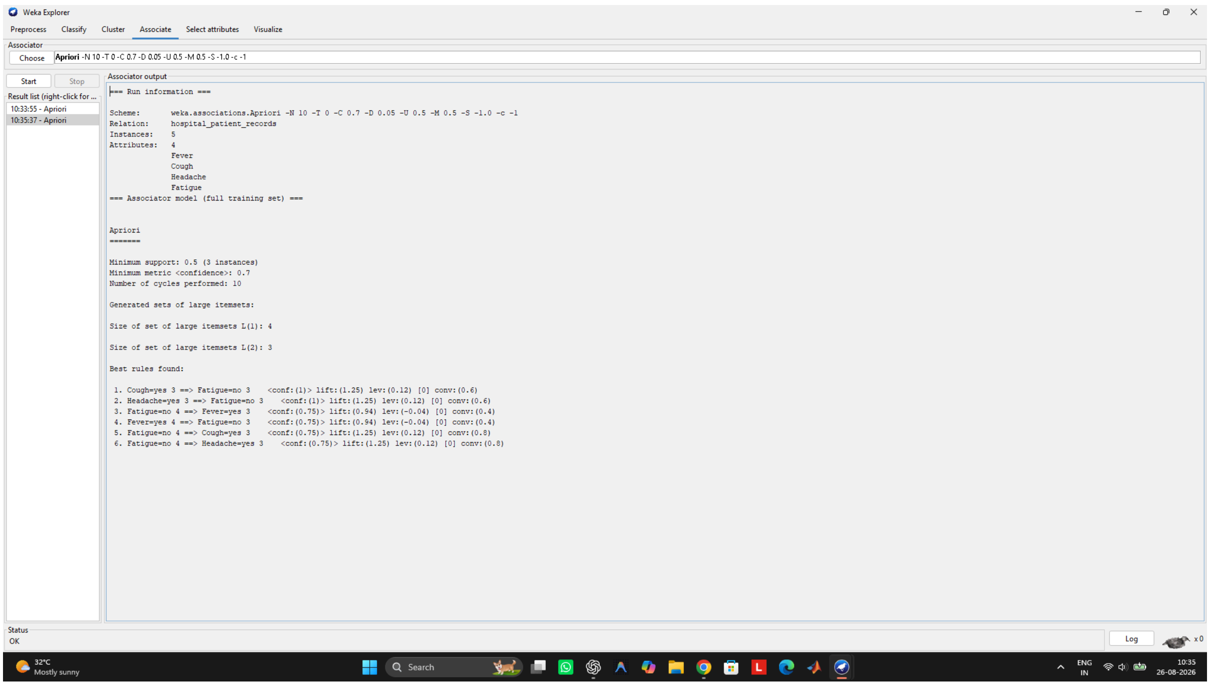
Therefore, the rule is a **valid constrained association rule**.

Interpretation

The rule indicates that among the patients having Fever, **75% also have no observed Fatigue** in this dataset. The support of 60% shows that the combination occurs in a significant portion of the patient records.

Conclusion

Apriori successfully identified a rule that satisfies the specified support, confidence, and Fever-based constraint. Constraint-based mining reduces irrelevant rules and focuses the analysis on patterns related to the required symptom.



Q2. Online Learning Platform

Objective

Apply the **Apriori algorithm** to identify frequent course combinations with the following constraint:

- Minimum Support = **40%**
- Each itemset must contain **at least two courses**

Dataset

Student	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

WEKA Implementation

The dataset was loaded into **WEKA Explorer → Associate → Apriori**.

The Apriori parameters were set with:

- Minimum Support = **0.40 (40%)**
- Output Itemsets = **True**

WEKA Output Screenshot

[Insert your WEKA Apriori screenshot here]

Figure 1: Apriori output for the Online Learning Platform dataset.

Frequent Itemsets

Since there are 5 students:

$$40\% = \frac{2}{5}$$

Therefore, an itemset must occur in at least **2 students** to be considered frequent.

1. Python + SQL

Appears in:

- S1
- S4

Support:

$$Support(ythonSQL) = \frac{2}{5} \times 100 = 40\%$$

Therefore, **{Python, SQL} is a frequent itemset**.

2. Python + Data Science

Appears in:

- S2
- S4

Support:

$$Support(ythonData\,Science) = \frac{2}{5}\times100 = 40\%$$

Therefore, **{Python, Data Science} is a frequent itemset**.

Non-Frequent Itemsets

The remaining two-course combinations occur only once or not at all, giving support below 40%.

The three-course combination:

Python + SQL + Data Science

appears only in S4:

$$Support = \frac{1}{5}\times100 = 20\%$$

Therefore, it is **not frequent**.

Constraint-Based Pruning

The constraint that every itemset must contain **at least two courses** eliminates all single-course itemsets from consideration. This reduces the number of candidate itemsets generated by Apriori and avoids unnecessary support calculations.

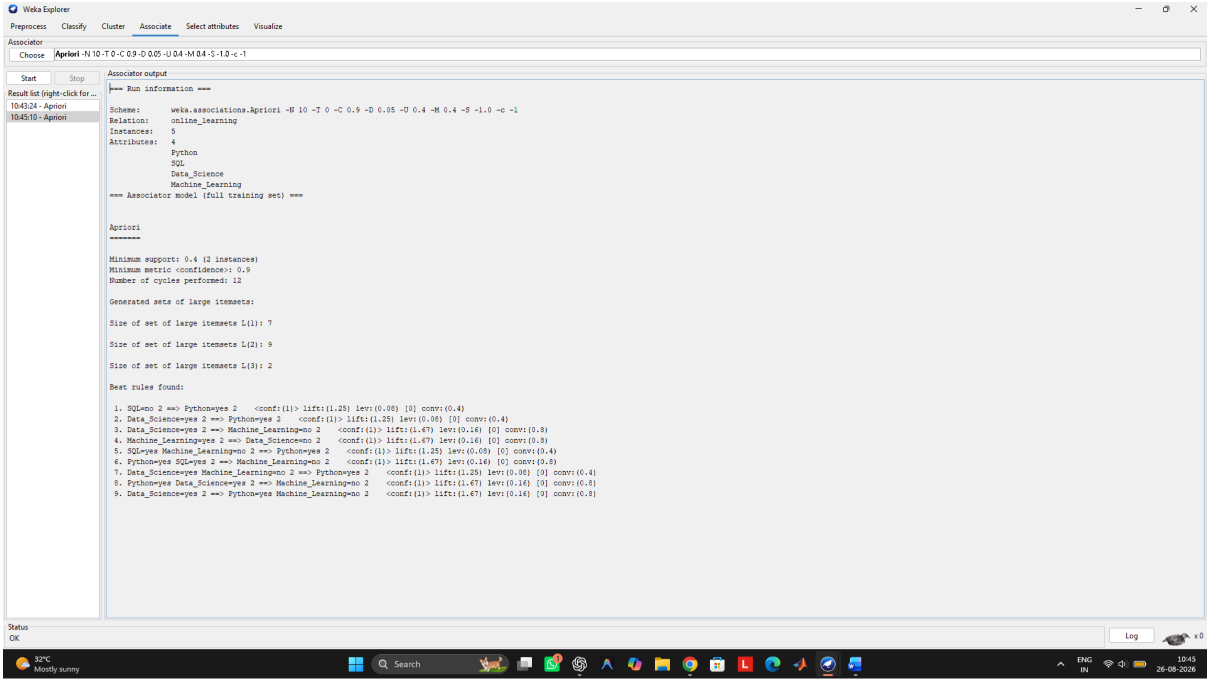
Final Result

The frequent itemsets satisfying the 40% minimum support and at-least-two-course constraint are:

Frequent Itemset	**Support**
Python, SQL	**40%**
Python, Data Science	**40%**

Conclusion

Constraint-based Apriori mining identifies meaningful course combinations while reducing unnecessary candidate generation. The results show that **Python and SQL** and **Python and Data Science** are the most frequent course combinations in the given dataset.



Q3. Library Borrowing Records

Objective

The Apriori algorithm is applied to identify frequent book combinations using a minimum support of **40%**. A constraint is imposed that the maximum itemset size should not exceed **3 items**.

WEKA Implementation

The dataset was loaded into **WEKA Explorer → Associate → Apriori**.

Parameters used:

- Minimum Support = **40%**
- Apriori Algorithm
- Itemset size considered up to **3**

WEKA Output Screenshot

[Insert your WEKA Apriori screenshot here]

Figure 1: Apriori output for the Library Borrowing Records dataset.

Frequent Itemsets

Data Mining + Python Programming

Occurs in T1 and T4:

$$Support = \frac{2}{5} \times 100 = 40\%$$

Data Mining + Database Systems

Occurs in T2 and T4:

$$Support = \frac{2}{5}\times 100 = 40\%$$

Python Programming + Database Systems

Occurs in T3 and T4:

$$Support = \frac{2}{5}\times 100 = 40\%$$

The three-itemset **Data Mining + Python Programming + Database Systems** occurs only in T4:

$$Support = \frac{1}{5}\times 100 = 20\%$$

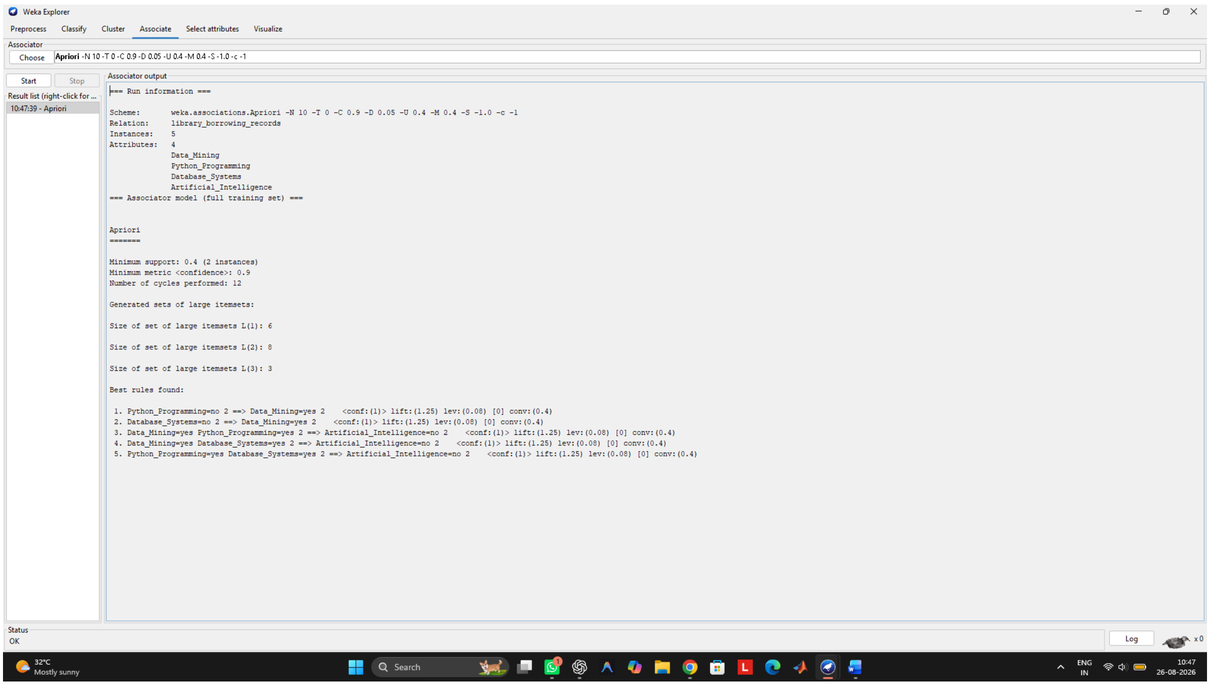
Therefore, it is not frequent.

Effect of the Constraint

Limiting the maximum itemset size to **3** prevents Apriori from generating itemsets containing four or more books. This reduces the number of candidate combinations, lowers memory requirements, and decreases the computational effort required during the candidate-generation process.

Conclusion

The frequent 2-itemsets are **Data Mining–Python Programming**, **Data Mining–Database Systems**, and **Python Programming–Database Systems**, each with 40% support. The maximum-size constraint makes the Apriori mining process more efficient by restricting unnecessary larger candidate itemsets.



Q4. University Course Hierarchy

Objective

Apply multilevel association rule mining with the constraint that rules should be generated only at the **Computer Science** level.

Dataset

Student	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

WEKA Implementation

The dataset was loaded into **WEKA Explorer → Associate → Apriori**.

The Apriori algorithm was used to identify frequent course associations. A minimum support of **40%** was considered for the analysis, and the association rules were examined specifically for the **Computer Science** level.

WEKA Output Screenshot

[Insert your Q4 WEKA Apriori screenshot here]

Figure 1: Apriori association-rule mining output for the University Course Hierarchy dataset.

Analysis

There are 5 students in the dataset.

Computer Science is taken by:

- S1
- S3
- S4

Therefore:

$$Support(Computer\ Science) = \frac{3}{5} \times 100 = 60\%$$

Machine Learning is taken by:

- S1
- S4

Therefore:

$$Support(Machine\ Learning) = \frac{2}{5} \times 100 = 40\%$$

Both **Computer Science and Machine Learning** occur together in S1 and S4:

$$Support(SML) = \frac{2}{5} \times 100 = 40\%$$

Association Rules

Rule 1: Machine Learning → Computer Science

$$Confidence(ML \rightarrow CS) = \frac{Support(LCS)}{Support(ML)} = \frac{2}{2} = 100\%$$

Therefore:

- Support = **40%**
- Confidence = **100%**

This is a strong association because every student taking Machine Learning in this dataset also takes Computer Science.

Rule 2: Computer Science → Machine Learning

$$Confidence(CS \rightarrow ML) = \frac{2}{3} = 66.67\%$$

Therefore:

- Support = **40%**
- Confidence = **66.67%**

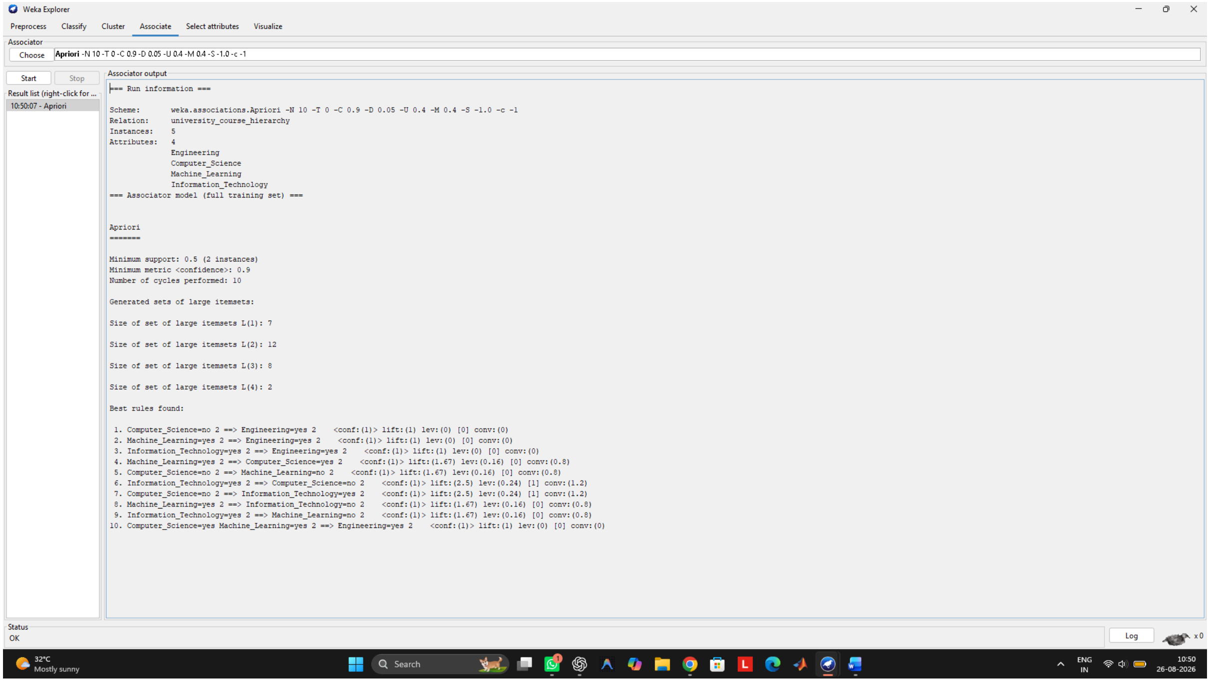
Effect of Course Hierarchy

The course hierarchy allows the mining process to focus on a specific level, in this case **Computer Science**, instead of generating and analysing all possible course relationships. This reduces irrelevant rules and makes the discovered associations easier to interpret.

The strong relationship between **Machine Learning and Computer Science** suggests that these courses are closely associated in the students' course selections.

Conclusion

Multilevel association rule mining successfully identifies meaningful relationships at the Computer Science level. The rule **Machine Learning → Computer Science** has **40% support and 100% confidence**, making it the strongest relevant association in this dataset. Such information can help universities understand course combinations and design suitable course pathways and recommendations.



Q5. Insurance Customer Records

Objective

Apply multidimensional association rule mining with the constraint **Age Group = Young**. The purpose is to identify relationships between income level and insurance policy purchased and use these patterns for targeted marketing.

Dataset

Customer	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Applying the Constraint

The constraint **Age Group = Young** filters the original five records to three customers:

Customer	Income Level	Policy Purchased
C1	High	Health Insurance
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

Thus, the filtered dataset contains **3 Young customers**.

WEKA Implementation

The dataset was loaded into **WEKA Explorer → Preprocess**. The constraint **Age Group = Young** was applied using an instance-value filter. The filtered dataset was then analyzed using **Associate → Apriori**.

The relevant Apriori settings were:

- Minimum Support = **33.33%**
- Minimum Confidence = **70%**
- Metric = **Confidence**

WEKA Output Screenshot

[Insert your WEKA screenshot here]

Figure 1: Apriori output after applying the Age Group = Young constraint.

Association Patterns

1. High Income → Health Insurance

Only C1 satisfies this combination.

$$Support = \frac{1}{3} \times 100 = 33.33\% Confidence = \frac{1}{1} \times 100 = 100\%$$

Therefore, this is a valid pattern.

2. Medium Income → Vehicle Insurance

Only C5 satisfies this combination.

$$Support = \frac{1}{3} \times 100 = 33.33\% Confidence = \frac{1}{1} \times 100 = 100\%$$

Therefore, this is a valid pattern.

3. Low Income → Travel Insurance

Only C3 satisfies this combination.

$$Support = \frac{1}{3} \times 100 = 33.33\% Confidence = \frac{1}{1} \times 100 = 100\%$$

Therefore, this is also a valid pattern.

Interpretation and Marketing Strategy

The constraint-based analysis shows different insurance preferences among Young customers based on income level.

- **High-income Young customers:** Target with premium Health Insurance plans, enhanced coverage, and additional health benefits.
- **Medium-income Young customers:** Promote affordable Vehicle Insurance packages and vehicle-related add-ons.

- **Low-income Young customers:** Offer budget-friendly Travel Insurance plans and flexible payment options.

This targeted approach can reduce irrelevant advertisements and improve customer engagement.

Conclusion

Multidimensional association rule mining successfully identifies income-policy relationships after restricting the analysis to **Young customers**. The patterns **High → Health Insurance**, **Medium → Vehicle Insurance**, and **Low → Travel Insurance** each have **33.33% support and 100% confidence** within the filtered dataset. These insights can support personalized marketing campaigns and improve the effectiveness of insurance product recommendations.